

Practical No. 7

Aim :

Introduction to No SQL using MongoDB with Aggregation and Indexing.

Objective :

- To understand the concept of No SQL database.
- To create & manage MongoDB collection.
- To perform CRUD (Create, Read, Update, Delete) operations.
- To use Aggregation Framework for data analytics.
- To create indexes and observe performance improvements.

Introduction to No SQL :

No SQL database are non-relational databases used for handling large volumes of unstructured or semi-structured data.

Introduction to MongoDB :

MongoDB is a popular NoSQL database that stores data in JSON-like documents (BSON Format). It is flexible, scalable & widely used in modern applications.

- Data Set $\rightarrow$ Collection $\rightarrow$ Document.

CRUD Operations :

1. Create - `insertone()`
`insertmany()`

2. Read - `Find()`

Findone()

3. Update - updateone()
updateMany()

4. Delete - ~~delete~~ deleteOne()
deleteMany()

Aggregation in MongoDB:

Aggregation is used to process data & return computed results (like sum, average).

Ex.,

```
db.users.aggregate ( [  
  { $ group: { _id: "$city", avgAge:   
    { $ avg: "$age" } } }  
])
```

Indexing in MongoDB:

Index improves the speed of queries by creating a fast lookup structure.

Ex.,

```
db.users.CreateIndex ( {"age" : 1 } )
```

Conclusion :

MangoDB provides efficient handling of large dataset using Flexible schema design. Aggregation helps in data analysis, while indexing improves query performance.

~~Ans~~
28/4/24